

A prepositional cycle in French

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Abstract

This paper discusses the morphosyntax and semantics of locative prepositions in French, with a particular focus on the prefix *de-* in complex prepositions (e.g., *en dessous de* [in from.under of] ‘below’) and in axial nominalizations (e.g., *le dessous de* [the from.under of] ‘the bottom part/region of’). We propose that in these formations, *de-*, which developed historically from an ablative marker, retains a Source semantics and lexicalizes the function PROJECT, which defines the region outward from the axial part of an object (Matushansky & Zwarts, 2019). We support our analysis with a corpus study on French child data showing that children acquiring L1 French tend to overuse *de-* in transitive contexts, overtly expressing conceptual structure that is typically implicit in the standard adult language, but replicating patterns present in Old and Medieval French.

Keywords: locative prepositions, ablative-locative transfer, axial nominalizations, child French.

1. Introduction

Like many other languages, French has morphologically simple and complex locative prepositions (Ruwet 1982, Zribi-Hertz 1984, Borillo 2001, Roy & Svenonius 2009). As shown in Table 1, locative prepositions differ in their transitivity, that is, whether they require an overt noun complement. Many prepositions notably also contain the prefix *de-*, which has been claimed to be an ablative Source preposition meaning ‘from’ in previous stages of French (Mackenzie 1978, Nikitina 2017).

Table 1. Simple, *de*-prefixed and multi-word locative prepositions in French

Simple P Transitive	<i>de</i>-prefixed P Intransitive (standard Fr.)	multi-word P Transitive
<i>en, dans</i> in, from.in ‘in’	<i>dedans</i> from.from.in ‘inside’	<i>au/en dedans de</i> at.the/in from.from.in of ‘at/in the inside of’
<i>sur</i> on ‘on’	<i>dessus</i> from.on ‘above’	<i>au/en/sur le dessus de</i> at.the/in/on from.on of ‘at the/in/on (the) top of’
<i>sous</i> under ‘under’	<i>dessous</i> from.under ‘below’	<i>en/au/sur le dessous de</i> in/at.the/on the from.under of ‘in/at the/on the bottom of’
—	<i>derrière</i> from.back [also transitive] ‘behind’	<i>au derrière de</i> at.the from.back of ‘at the backside of’
—	<i>devant</i> from.front [also transitive] ‘in front, ahead’	<i>au devant de</i> at.the from.front of ‘in front of, ahead of’
<i>hors</i> (archaic) out ‘out’	<i>dehors</i> from.out outside	<i>en/au dehors de, hors de</i> in/at.the from.out of, out of ‘in/at the outside of, out of’

In this paper, we provide evidence for the idea that *de-* remains an ablative marker in locative formations in modern French. Ablative markers are incorporated into the morphological makeup of complex locative prepositions in many other languages; this is known as ‘ablative-locative’ transfer (Mackenzie 1978, Nikitina 2017, a.o.). The presence of a Source morpheme in locative prepositions may at first sight be surprising, since it is at odds with the locative semantics of the complex prepositions at hand (cf. the ‘locative ablative dichotomy’ in Anderson’s 1971 theory of case).¹ Examples from other languages of prepositions with a locative meaning despite the presence of an ablative morpheme in their morphosyntactic make-up are given in (1); ablative morphemes are shown in bold.

¹ As Nikitina (2017) observes, there are many fewer locative prepositions with an allative (Goal) morpheme, which is rather surprising given the frequent syncretism across languages between locative and allative prepositions (e.g., French *à* ‘at/to’).

- (1) a. *in.tus* (in.from) ‘in’, *de.super* (from.above) ‘above’ (classical Latin)
 b. *d.entro* (from.in) ‘in’ (Spanish, Italian)
 c. *mi-taxat* (from-under) ‘under’ (Hebrew)

The addition of an ablative morpheme in prepositions with a locative meaning can in fact take place iteratively in the same language (MacKenzie 1978, Nikitina 2017). The diachronic development of French (*de*) *dedans* ‘(from) inside’ from Classical Latin *intus* ‘inside’ is a good example of such iterative patterns, as shown in (2); see Rainsford (2019), Amiot et al. (2020: 876), and Troberg (2021) on the diachrony of *de*-prefixed prepositions.²

- (2) a. *in.tus* ‘from inside’ (source meaning, classical Latin)
 b. *in.tus* ‘inside’ (locative meaning, classical Latin)
 c. *de.intus* ‘from inside’ (source meaning, vulgar Latin)
 d. *de.intus* ‘inside’ (locative meaning, vulgar Latin)
 e. *d.en.z* ‘inside’ (locative meaning, Old French)
 f. *de.d.en.z* ‘inside’ (locative meaning,³ Old French)
 g. *d.an.s* ‘inside’ (locative meaning, Modern French)
 h. *de.d.an.s* ‘inside’ (locative meaning,⁴ Modern French)
 i. *de de.d.an.s* ‘from inside’ (source meaning, Modern French)

As one can see from (2), the addition of the ablative morpheme to a preposition with a locative meaning can have two different effects. It can either endow the resulting P-formation with a source meaning (see e.g., (2h)-(2i)), or is semantically inert and leaves the original locative meaning intact ((2e)-(2f)). The question is then to identify the force behind the latter cyclical change: why is an ablative morpheme co-opted by a P with locative semantics, with no apparent semantic effect?

1.1. *de*-prefixed locative prepositions

Among the languages which exhibit ablative-locative transfer, French exhibits two interesting properties. The first is that for all prepositional doublets that alternate between a

² In Latin *intus*, the *-(i)tus* morpheme is an adverbializer with an ablative/separative meaning (Rosén 2007; see e.g., *originitus* ‘from their origin’ or *funditus* ‘from the bottom’). In the Old French *denz* formation, *-z* (or the orthographic variant *-s*) is the fossilized trace of this Latin morpheme, as is *-s* in Modern French.

³ While Nikitina (2017: 74) argues that *dedenz* is ambiguous between a source and a locative meaning in Old French, we were not able to find any occurrences of this preposition with a source meaning in the FranText Old French subcorpus. We thus take *dedenz* to only have a locative meaning.

⁴ Note that Modern French can also embed a locative P headed by *dans* ‘in’ within a Source P headed by *de* ‘from’. The resulting formation (where *de* and *dans* must be separated by a blank according to orthographic conventions) has a source meaning; see (i) for some examples taken from the French FrTenTen23 corpus from Sketch Engine (Kilgariff et al. 2014).

- (i) a. nous ne pouvions plus sortir de dans la boue
 we NEG could anymore go out from in the mud
 ‘We couldn’t get out of the mud anymore.’ (FrTenTen23)
 b. Le Lieutenant sortit de dans les feuillages.
 the lieutenant came from inside the leaves
 ‘The lieutenant came from within the leaves.’ (FrTenTen23)

simple preposition and a *de*-prefixed counterpart (e.g., *dans* ‘in’ and *dedans* ‘inside’), the variant without *de*- is strictly transitive, while the variant with *de*- is strictly intransitive in standard French. As illustrated in (3)-(4), simple Ps require an overt noun complement, while *de*-prefixed Ps reject them.

(3) Simple Ps are transitive

- a. Il est dans *(le vase).
it is in the vase
- b. Elle est sur *(la table).
it is on the table
- c. C’est sous *(le lit).
it is under the bed

(4) *de*-prefixed Ps are intransitive

- a. Il est **dedans** (%le vase).
it is from.in the vase
- b. Elle est **dessus** (%la table).
it is from.top the table
- c. C’est **dessous** (%le lit).
it is from.bottom the bed

In regional, colloquial or archaic varieties, however, *de*-prefixed locative Ps have been reported to appear with an overt complement (Frei 1929, Hagège 1997: 23, Ruwet 1982, Zribi-Hertz 1986: 601, Abeillé & Godard 2021: 792). as illustrated in (5); the symbol ‘%’ in (4)-(5) indicates availability in non-standard variants, following the convention adopted in Abeillé & Godard (2021).

- (5) a. %Ils sont cachés dessous les ailes. (non-standard variant)
they are hidden from.bottom the wings
‘They are hidden underneath the wings.’
- b. Ils sont cachés sous les ailes. (standard variant)
they are hidden bottom the wings
‘They are hidden under the wings.’

The use of the *de*-prefixed locative Ps which are strictly intransitive in standard French in transitive, simple P contexts was also present in earlier historical stages of the language (Rainsford 2019, Amiot et al. 2020, Troberg 2021), as shown in (6) from Medieval French.

- (6) Desoz le col Tristan a mis son braz.
from.under the neck Tristan has put his arm
‘Tristan put his arm under the neck.’ (Bérout, *Le Roman de Tristan*, 1170)

Similar non-target transitive uses of *de*-prefixed Ps have been reported in child French. As illustrated in (7), child L1 French learners also sometimes use *de*-prefixed locative Ps with an overt complement (Pallaud & Savelli 2001).

- (7) je veux pas laver dedans la cuvette.
I want not wash from.in the basin
‘I don’t want to wash in the basin.’ (Madeleine, 2;09, Paris Corpus, CHILDES)

We suggest that children’s overuse of *de*-prefixed locative Ps in simple P contexts recalls previous historical stages of the language. French children seem to be returning to a prior stage of a sort of prepositional cycle, where locative Ps are built with an ablative

marker.⁵ The non-target addition of *de-* can thus be seen as a tendency for the ablative marker to re-enter the cycle of change, and to overtly express whatever is conveyed by the ablative case marker in locative Ps.

1.2. *de*-prefixed axial nominalizations

A second interesting property of French among languages with locative-ablative transfer is that the *de-* prefix is often retained in axial nominalizations—DPs that are morphologically related to the simple locative P, as illustrated in (8). Axial nominalizations are derived from axial morphemes (Svenonius 2006), such as *sur* ‘on’ in French, and are used to name parts of an object.⁶

- (8) le dessous/ *sous de la maison (French)
 the from.under/ under of the house
 ‘the bottom part/region of the house’

Standard Modern Hebrew similarly has locative-ablative transfer. However, the ablative morpheme *mi-* cannot appear in axial nominalizations in the language, as shown in (9) (Itai Bassi, p.c.).

- (9) taxtit/ *mi-taxtit ha-bayit. (Hebrew)
 bottom/ from-bottom the-house
 ‘the bottom part of the house’

This paper addresses the following research questions:

- What is the role of *de-* in complex Ps in French? Why does an ablative marker often appear in the morphological make-up of locative Ps across languages?
- Why is *de-* banned in transitive Ps in standard French? Why does *de-* occasionally surface in transitive Ps in child French and other non-standard varieties of French?
- What is the semantic difference, if any, between derived DPs built with or without an ablative marker (e.g., between Hebrew and French DPs in (8) and (9))?

Regarding the first question and elaborating on a suggestion from Matushansky & Zwarts (2019), we shall argue that the ablative morpheme regularly found in complex locative prepositions across languages introduces *project*, a function that takes an axis of the object of reference (e.g., its front) and returns the set of vectors external to the object that point in the same direction as the axis (e.g., the region in front of the car). This echoes a previous idea by Nikitina (2017), who proposes to account for the presence of ablative morphemes in locative Ps by the fact that the Figure is located *from* the object of reference: in order to locate the Figure, one (mentally) starts from the object of reference and (mentally) goes towards a direction *from* this departure point. For Nikitina, the ‘path’

⁵ See Waters (2009) on the related idea that English prepositions go through a cycle through which a new spatial element is periodically added.

⁶ As our translation of (8) already suggests, we will argue that this is not the only meaning available for *de*-prefixed nominalizations.

expressed by the ablative in locative P is a fictive, mental path—what Talmy (1996) calls an ‘access path’.

The paper is organized as follows. In section 2, we discuss the syntax and semantics of simple and complex locative prepositions. We then focus on the role of the prefix *de-*, first in axial nominalizations in section 3, and then in complex prepositions in section 4. Section 5 investigates the non-target transitive use of *de*-prefixed Ps in child French, and then discusses ways to account for this trend.

2. Syntax and semantics of French locative prepositions

Following Svenonius (2006, 2010) and Roy & Svenonius (2009), we assume that morphologically simple and complex locative prepositions can be decomposed into the same complex conceptual structure involving multiple projections. Morphologically simple and complex Ps have mostly the same syntax/semantics; simple Ps leave much of this shared conceptual structure unrealized. As Roy & Svenonius (2009) note, this view was prefigured by Ruwet (1982: 318), who proposed that simple Ps like *sur* ‘on’ are underlyingly complex Ps (e.g., *à dessus* [at from.on] in the case of *sur*).

On this view, the (a) and (b) sentences in (10)–(11), for instance, which only differ in the morphological complexity of the preposition, roughly convey the same truth-conditional meaning (a view explicitly endorsed by Ruwet 1982).

- (10) a. Il se cache sous la table.
He REFL hides under the table
‘He’s hiding under the table.’
b. Il se cache en dessous de la table.
He REFL hides in from.under of the table
‘He’s hiding underneath the table.’
- (11) a. Il est dans le cratère.
it is in the crater
‘It is in the crater.’
b. Il est au dedans du cratère.
it is at.the from.in of the crater
‘It is inside the crater.’

Meaning differences, if they exist, are in general subtle and non-at-issue.⁷ We suggest that these meaning differences may be derived as manner implicatures via the *Be Brief* submaxim of manner (Grice 1975, Rett 2015, Levinson 2000). The reasoning proceeds as follows: if the speaker chooses the marked form out of two truth-conditionally equivalent expressions, it is because they are after a marked meaning. In the case of locative prepositions, the marked meaning seems to be (at least in some cases) a contrastive emphasis between the asserted spatial configuration and alternative spatial configurations.

⁷ See section 4 on the apparent counter-example of supressive Ps, which often require contact between the Figure and the object of reference when they are morphologically simple.

For instance, (14b) is rather odd in French, because via the choice of the more complex preposition, the speaker implicates that Tom could be above the shower rather than under it.⁸

- (14) a. Tom est sous la douche.
Tom is under the shower
'Tom is in [under] the shower.'
- b. #Tom est en dessous de la douche.
Tom is in from.under of the shower
'Tom is underneath the shower.'

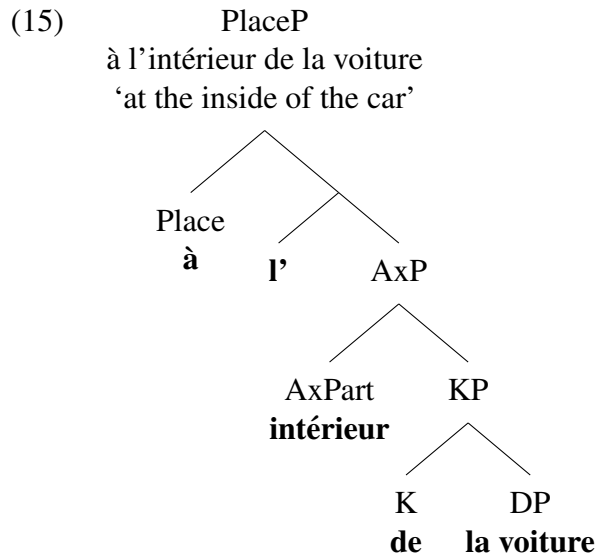
The syntax and semantics we adopt for the prepositions under study largely builds on Matushansky & Zwarts (2019), Svenonius (2006, 2010) and Roy & Svenonius (2009). (15) provides the structure proposed by Roy & Svenonius (2009) for *à l'intérieur de la voiture* 'at the inside of the car', which includes a case projection notated 'K' and an axial part projection notated AxPart (Svenonius 2006).

⁸ In contrast with (14b), *Le chat est en dessous de l'armoire* 'The cat is below the cupboard' is felicitous because it is plausible to implicate that the cat could be above the cupboard instead of below it; see Vandeloise (1986a) for a similar proposal.

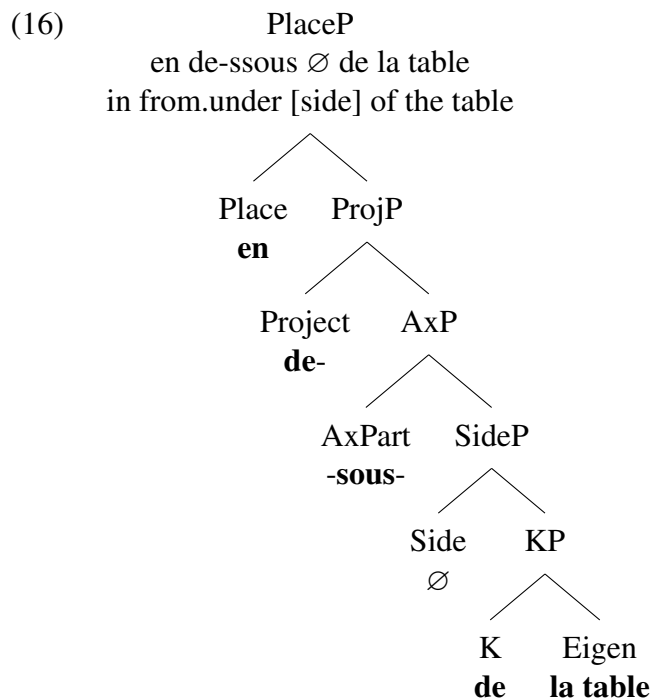
By assuming the same semantic structure for simple and complex Ps when they are used transitively, we do not necessarily commit to the view that simple Ps used transitively and the (necessarily complex) counterpart used intransitively appear in the exact same contexts. The change in argument structure can indeed very well trigger distinct semantic/pragmatic effects that may be responsible for the often observed fact that Ps without an overt complement have more restricted uses than their (simple) counterparts with an overt complement (Tuller 1991: 42 and Authier 2016: section 3 on French). These restrictions are shown for *dans/dedans* in (12) and for *sur/dessus* in (13) (contra Authier 2016: 257, who argues that such restrictions do not hold for *dessus*).

- (12) a. Il vend ses crêpes dans les rues de Paris.
he sells his crêpes in the streets of Paris
'He sells his crêpes in the streets of Paris.' (Authier 2016: 254)
- b. #Les rues de Paris, il vend ses crêpes dedans.
the streets of Paris he sells his crêpes in [it]
Intended: 'The streets of Paris, he sells his crêpes in them.' (Authier 2016: 254)
- (13) a. Je suis sur Paris.
I am on Paris
'I'm in Paris.'
- b. #Paris, je suis dessus.
Paris, I am on [it]
Intended: 'Paris, I'm in it.' [acceptable only if 'Paris' is understood as referring to e.g., a project related to Paris]

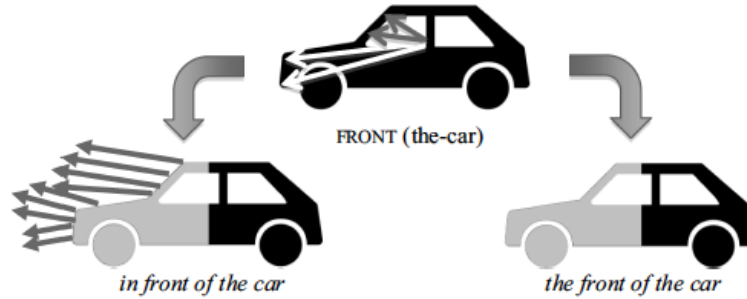
Authier (2016) accounts for the above contrast in the semantics, by positing that one of the meanings of the simple P *dans* is not expressible by the complex P *dedans*. However, it seems to us that a pragmatic explanation in terms of contrastive inference can account for these asymmetries more economically: in (12b) and (13b), the complex P is in sentence-final position, which is the canonical position for contrastive focus. As a result, the assertion evokes alternative spatial configurations, which make little sense here (the seller can only sell his crêpes *in* the streets of Paris; and *être sur Paris* in the intended meaning does not have clear alternatives; for instance, #*Je suis SUR Paris, pas DEDANS* sounds nonsensical).



To the above structure we add a projection called Project, and split the above AxPart in two different projections AxP and SideP, for reasons to be made clear presently. The resulting proposed hierarchy of projections is given in (16), illustrated with *en dessous de* 'underneath'. Importantly, we assume the same structure for morphologically simple and complex locative prepositions.



The axial morpheme (e.g., *sous* in (16)) is the core element in locative prepositions. While Svenonius argues that axial parts are functional elements, Matushansky & Zwarts (2019) offer arguments in favor of an analysis in terms of weak nouns (gender agreement, number, case, and the intimate relation with nominalizations referring to parts of an object, such as *le dessous* 'the bottom/underside' in French). Axial nominals like

Figure 1. Axial derivatives (Figure taken from Matushansky & Zwarts 2019: 274)

sous ‘under’ are interpreted with respect to an axis. The axis determines the meaning of both the derived axial nominalization, which denotes a *part of an object* (e.g., *l’avant de la table* ‘the front of the table’) as well as the derived locative preposition (e.g., *au devant de la table* ‘at the front part/region of the table’), which denotes a *spatial region*.

Following Zwarts & Winter (2000), Matushansky & Zwarts (2019) model axes as sets of vectors pointing from the center of an entity towards a specific direction. For instance, the axial nominal *front* is defined as in (17) below, that is, as a function that takes as its input an entity x and returns the set of vectors u starting at the middle point of x and ending at the boundary of x with a forward direction (with respect to a certain frame of reference). The vectors returned by *front* are graphically represented by the middle picture in Figure 1 (taken from Matushansky & Zwarts 2019).⁹

$$(17) \quad \llbracket \text{front} \rrbracket = \lambda x \lambda u. \text{start}(u) = \text{center}_c(x) \wedge \text{END}(u) \in \text{boundary}(x) \wedge \text{forward}(u)$$

(Matushansky & Zwarts 2019: 274)

From the axial nominal, Matushansky & Zwarts (2019) derive both the *object part* meaning, always active in the derived nominalization (e.g., *the front of the car*, see the right-side picture on Figure 1), as well as the *projection* meaning, active in the preposition (e.g., *in front of the car*, see the left-side picture on Figure 1).

The object part meaning involves the reifying function *object*. This function maps an axis to the unique object occupying it, as in (18). This allows Matushansky & Zwarts (2019) to account straightforwardly for the use of the definite with axial nominalizations.

$$(18) \quad \llbracket \text{front}_{\text{part}} \rrbracket = \lambda x. \text{object}(\text{front}(x))$$

(the front part of x)
(Matushansky & Zwarts 2019: 275)

The projection meaning involves the function *project*. The job of *project* is to define the region outward the axial part of the object, e.g., the region defined by *devant la voiture* ‘in front of the car’ (as graphically represented by the vectors in the left picture on Figure 1). It does so by projecting the relevant internal axis, that is, the set of vectors

⁹ As an anonymous reviewer points out, the point at which the vectors defining the axis start may be more accurately described as a contextually determined central point of the entity (what we express as center_c in (17)) than its objective center. Take, for instance, an auditorium with two seating sections divided by an aisle, with the front section occupying 40% of the space and the back section occupying 60%. As the reviewer notices, *the front of the auditorium* and *the back of the auditorium* would not be construed relative to the objective geographical center of the space, but rather relative to this dividing aisle.

internal to the object (defined by the axial nominal *front*) to the set of vectors *external* to the object that point in the same direction as the axis, as in (19)-(??). Note that following the definition in (??), the ‘project’-vectors start where the ‘axis’-vectors end (as also graphically represented in Figure 1).

(19) $\llbracket \text{front}_{\text{proj}} \rrbracket = \lambda x. \text{project}(\text{front}(x))$ (a region projected from the front of x)
(Matushansky & Zwarts 2019: 275)

(20) $\llbracket \text{project} \rrbracket = \lambda f \lambda u. \exists w [f(w) \wedge \text{start}(u) = \text{end}(w) \wedge \text{dir}(u) = \text{dir}(w)]$
(Matushansky & Zwarts 2019: 275)

ProjP is not part of Svenonius’ inventory of projections but naturally derives from the semantics of locative prepositions just summarized. In the specifier of ProjP, we find the morpheme spelling out the function *project*. Matushansky & Zwarts (2019: 278) suggest that the ablative morpheme regularly found in locative prepositions across languages lexicalizes *project*. We take up this suggestion and propose that in French, *de-* spells out *project*; we explore this idea further in the next sections.¹⁰

Following up on a suggestion by Svenonius (2010: 131), we propose to decompose prepositions like English *inside* in two different elements *in-* and *-side*.¹¹ While the axial morpheme must be spelled-out overtly both in simple and complex prepositions because it is central to their meaning, the abstract SIDE noun can remain silent, as its contribution is highly predictable. For instance, English has both *inside* and *in* (but not *side* for obvious reasons). Abstract SIDE-nouns are regularly found cross-linguistically in the morphosyntactic make-up of locative prepositions or affixes. While their presence is particularly transparent in English *inside*, *alongside*, or *outside*, such morphemes are also present in a more opaque way in locative prepositions of Romance languages. For instance, the *-tr-* and *-pr-* morphemes found in the Italian prepositions *dietro* ‘behind’, *dentro* ‘inside’, *sopra* ‘above’ or French prefix *intra-* are etymologically related to Latin *parte*. *Dentro* ‘inside’ is derived from ablative *de-* and *intra*, itself derived from *intera*

¹⁰ The idea that *de* has Source semantics in these formations contrasts with the common view that morphologically simple French Ps never have Path semantics in French and in other Romance languages. Folli & Ramchand (2005), for instance, takes Italian *a*-PPs to be strictly locative, and any directional meaning comes only from the verb (see also Fábregas 2007 on Spanish and Acedo-Matellán 2016 on Latin). While it may be true that *de*-PPs do not introduce a path when used with a motion verb, we note that stative construals such as (21) below clearly convey a Source semantics that cannot come from the stative verb employed. We take this to show that *de* can introduce Source semantics on its own, at least in combination with stative predicates, as in the sentences we focus on in this paper.

- (21) a. Je suis de Bruxelles.
I am from Brussels
‘I am from Brussels.’
b. Il est rouge de la tête aux pieds.
he is red from the head to the toes
‘He’s red from head to toes.’

¹¹ We choose to call the corresponding projection Side rather than Part because the latter comes with a reifying flavour that should be avoided since axial ‘parts’ are in fact not object parts.

parte ‘internal part’, i.e. the ablative singular feminine of adjectives, modifying the noun *parte*; *sopra* derives from Latin *supera parte* ‘upper part’, and so on (Mackenzie 1978: 142). Interestingly, the meaning effect triggered by the overt use of the SIDE noun when it is optional (as in *in/inside*) is similar across languages. For instance, it can signal both in Italian and English that the target use of the Ground is the ‘container’ interpretation; in (22), for example, the tree is understood as the trunk rather than its crown, since the trunk, but not the crown, has sides (Tom Roberts, p.c; see also Svenonius 2010: 156).

- (22) a. I’m in the tree, but I’m not inside the tree. (English)
 b. Sono nell’albero, ma non dentro l’albero. (Italian)
 am in the tree but not inside the tree
 ‘I’m in the tree, not inside the tree.’

At the bottom of the tree in (16), we find the Eigen projection, which gives the *Eigenspace* of an entity, that is, the spatial region occupied by the referent. The function Eigen is never spelled out overtly across languages (Wunderlich 1991: 598).¹² K encodes the part/whole relation between the Eigenspace of the entity and the axis or axial side. It is either spelled out by *de*, which is the dedicated genitive case marker in French, or left zero (in simple P) (see also Franco 2019 on Italian locative prepositions).

Finally, the ‘outer’ preposition is housed in the Place projection. We assume that in the morphological makeup of complex prepositions, the outer preposition has roughly the meaning it has in other uses, that is, contiguity or overlap for *à* ‘at’, containment for *en* ‘en’, etc.¹³

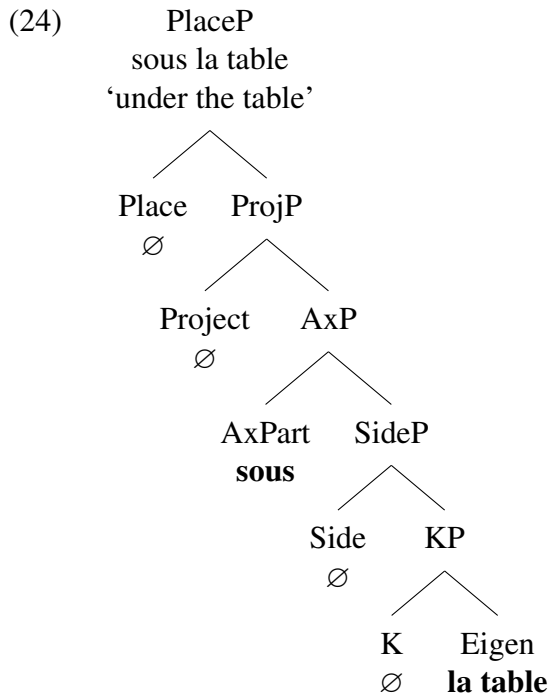
We assume that morphologically simple Ps have the same functional structure as complex ones. All projections in simple Ps except AxPart remain silent, as in (24).

¹² Svenonius (2006) (see also Svenonius 2010: 156, fn 6) speculates that genitive case (his K) is an overt expression of Eigen across languages. This seems rather implausible from a semantic perspective, and we therefore keep Eigen and K distinct.

¹³ See Matushansky & Zwarts 2019: section 6 for an alternative analysis of the outer preposition. The following minimal pair illustrates the fact that the outer preposition keeps the meaning it has in other uses:

- (23) a. Le château est au dessous du village.
 the castle is at.the from.bottom of.the village
 ‘The castle is below the village.’
 b. Le château est en dessous du village.
 the castle is inside from.bottom of.the village
 ‘The castle is underneath the village.’

While (23a) expresses a normal situation where the castle is situated in a region contiguous or overlapping with the bottom part of the village, (23b) weirdly suggests that it is contained in a region projected from the bottom of the village (in a big cave, for instance).



Equipped with this analysis, we can now turn to our research questions. In section 3, we address the role of *de-* in axial nominalizations, which we suggest are embedded in multi-word Ps with definite articles, as in (25).

- (25) a. au- dessus de
 at.the from.above of
 ‘above’
- b. au- dedans de
 at.the from.from.in of
 ‘inside’

We then examine *de-* in spatial prepositions in section 4. Our discussion focuses on how French compares with other languages with ablative-locative transfer.

3. *de-* in axial nominalizations

3.1. Axial nominalizations

French differs from other languages with ablative-locative transfer such as Hebrew in the morphological makeup of their axial nominalizations: while ablative morphemes are obligatory in many nominalizations derived from simple Ps in French, as in (26), they cannot appear in the corresponding nouns in standard Modern Hebrew, as in (27). The Hebrew data presented in this section were provided by Itai Bassi (p.c.).

(26) French axial nominalizations

- a. le *(de)ssous de la table
the from.under of the table
'the bottom part/region of the table'
- b. le *(de)ssus de la chaise
the from.above of the chair
'the upper part/region of the chair'
- c. le *(de)dans de la boîte
the from.from.in of the box
'the inside part/region of the box'

(27) Hebrew axial nominalizations

- a. ha-(*mi)-taxtit Sel ha-Sulxan
the from bottom of the table
'the bottom (part) of the table'
- b. ha-xelek ha-(*mi)-elyon Sel ha-kise
the part the from top of the chair
'the top (part) of the chair'

A key question is whether this morphosyntactic difference has semantic correlates. We claim that it does: while Hebrew axial nominalizations can only refer to a specific *part* of an object (e.g. its top, bottom), French ones can additionally be used to describe the (reified) *region projected from the axis* spelled out by the axial morpheme (e.g., the region above, below), if they are prefixed with *de-*. We propose that this is because *de-* lexicalizes *project*. We therefore expect a semantic difference, in French, between axial nominalizations prefixed with *de-* or not (e.g., between *le devant* 'the from-front' and *l'avant* 'the front'); only the former should be able to describe an entity external to the object. Furthermore, we expect the same kind of semantic difference to arise between French *de-*prefixed nominalizations and Hebrew axial nominalizations, which do not involve the ablative morpheme lexicalizing *project* in their morphological make-up.¹⁴

These two predictions are borne out. The standard Hebrew example in (29), without a Source P, can only denote an object part of the reference's object, while its French

¹⁴ There is a non-standard way to form axial nominalizations in Hebrew, illustrated in (i). The adverbs *lemala* and *axora*, which in their basic use are Path morphemes meaning 'upwards/backwards' (and are morphologically complex), can be used in axial nominalizations to mean 'the top/back part of'. In these non-standard formations, the ablative morpheme is acceptable. We would therefore predict that that these variants can also denote a region projected from the axial part of the object of reference, as in French.

- (i) a. %ha-(mi)-lemala Sel ha-kise
the from upwards of the-chair
'the top of the chair'
- b. %ha-(mi)-axora Sel ha-masait
the from backwards of the truck
'the back of the truck'

counterpart in (28), with a Source P, can also describe the region projected from the bottom of the reference's object. Examples (28)-(29) are chosen such that they can only make sense if the axial nominalization in subject position describes a region below the table (and not the bottom part of the table).

(28) French

Le dessous de la table est totalement utilisable pour le stockage.
the from.under of the table is totally usable for the storage

'The region under the table can totally be used as storage space.' (Internet)

(29) Hebrew

#ha-taxtit Sel ha-Sulxan yaxol leSameS ke-mekom ixsun.
the.bottom of the-table can be.used as-place storage

'#The bottom (part) of the table can be used as a storage space.'

The ablative morpheme *de-* is critical for this projective meaning in French. Replacing *le dessous* 'the from.under' with *le fond* 'the bottom' in (30) is infelicitous, as it is missing *de-*, which contributes the projective meaning in axial formations.

(30) French

#Le fond de la table est totalement utilisable pour le stockage.
the bottom of the table is totally usable for the storage

'#The bottom (part) of the table can totally be used as storage space.'

In informal registers, Hebrew does have a felicitous counterpart of the French example in (28). In line with our hypothesis, however, the projective meaning requires the ablative morpheme *-mi-*; the nominal *taxtit* 'bottom' must furthermore be replaced with the preposition *taxat* 'under'.

(31) Hebrew

%ha-mi-taxat Sel ha-Sulxan yaxol leSameS ke-mekom ixsun.
the-from-under of the-table can be.used as-place storage

'The region/part under the table can be used as a storage place.'

French also has pairs of axial nominalizations prefixed either with *de-* or another preposition, namely *a-*. One of these pairs is *le devant* 'the from.front' and *l'avant* 'the front'. Thus for instance, while (32) is felicitous as a description of Pierre being located in the region projected from the front of the house, this is not the case in (33).¹⁵

¹⁵ According to Vandeloise (1986b: 286–287), axial nominalizations can never refer to outward regions in French, independently of their morphosyntactic makeup. For him, the possibility of referring to a region external to the house in examples such as (32) is due to the polysemy of nouns like *maison* 'house', which can also be used to refer to parts adjacent to houses, like gardens. However, this account predicts that examples like (33) should also be felicitous, contrary to fact.

- (32) Pierre se trouve sur le devant de la maison.
 Pierre REFL finds on the from-front of the house
 ‘Pierre is on the front/within the region in front of the house.’
- (33) Une fissure/ #Pierre se trouve sur l’avant de la maison.
 A crack/ Pierre REFL finds on the front of the house
 ‘A crack/Pierre can be found on the front of the house.’

Similarly, while example (34) suggests that the police are in the procession, in (35) the police are not part of the procession but in the region ahead of it.

- (34) La police se trouve à l’avant du cortège.
 the police REFL finds at the front of the procession
 ‘The police are at the front of the procession.’
- (35) La police se trouve au devant du cortège.
 the police REFL finds at the from front of the procession
 ‘The police are in front of the procession.’

de-prefixed axial nominalizations can thus refer to either object parts or regions projected from those parts, while axial formations without an ablative morpheme may only refer to the parts themselves.

3.2. Analysis

Based on these data, we argue that *de*-lexicalizes *project* in axial nominalizations. However, *project* cannot be the only function active in these formations, since they do not literally refer to spatial regions like prepositions. Rather, they refer to what we called above ‘reified regions’, namely, portions of the world occupying the spatial regions returned by the *de*-prefixed formation. The contrast in (36) helps to make this point clear.

- (36) a. Je lave/ #suis le dessous de la table.
 I wash am the from-under of the table
 ‘I’m washing/ am the region under the table.’
- b. Je lave/ suis dessous la table.
 I wash am from-under the table
 ‘I’m washing/ am under the table.’

While both examples can describe situations where the speaker is situated under the table washing the portion of the world situated below it (e.g., the ground, anything between the ground and the bottom of the table), only (36b) can be used as a description of my location. In (36a), the variant with the verb *être* ‘be’ is strange because it literally states an identity between me and the (reified) region under the table (the portion of the world occupying this region). We propose that this is because the function *object* always applies in axial nominalizations (Matushansky & Zwarts 2019), and returns the unique

entity occupying the region projected from the axial part of the entity via **project**, as in (37). Departing on this point from Matushansky & Zwarts, we therefore assume that **project** and **object** can be both active in the same formation.

$$(37) \quad \llbracket \text{dessous}_N \rrbracket = \lambda x. \text{object}(\text{project}(\text{bottom}(x)))$$

Another point of clarification concerns the object part use of axial nominalizations. While French *de*-prefixed axial nominalizations can indeed refer to the region returned by the *de*-form (as it is reified by **object**), they can also be used to refer to specific parts of an object. For instance, *le dessous de la table* ‘the from.under of the table’ can not only refer to the portion of the world projected from the bottom of the table, but also to the bottom of the table itself. We can derive the latter object part meaning of axial nominalizations in different ways. One option, shown in (38a), is to stipulate that in axial nominalizations, *de*- is polysemous and can either spell out **project** or **object**, the latter meaning being selected in the object part use of axial nominalization. A second option, shown in (38b), assumes that *de*- invariably has ablative semantics, while **object** remains covert in the nominalization, much like in (37). This would require us to define another version of **project** which does not necessarily return outward vectors with respect to the object. We define **project**₂ in (39b) to return a set of vectors whose starting point is equal to the starting point of vectors in the axial nominal. Unlike the vectors expressed by the axis, these **project**₂-vectors do not necessarily end at the boundary of *x*. But in the object part meaning, the vectors do end there, so that **object** ends up reifying the same region as it does in the absence of **project**.

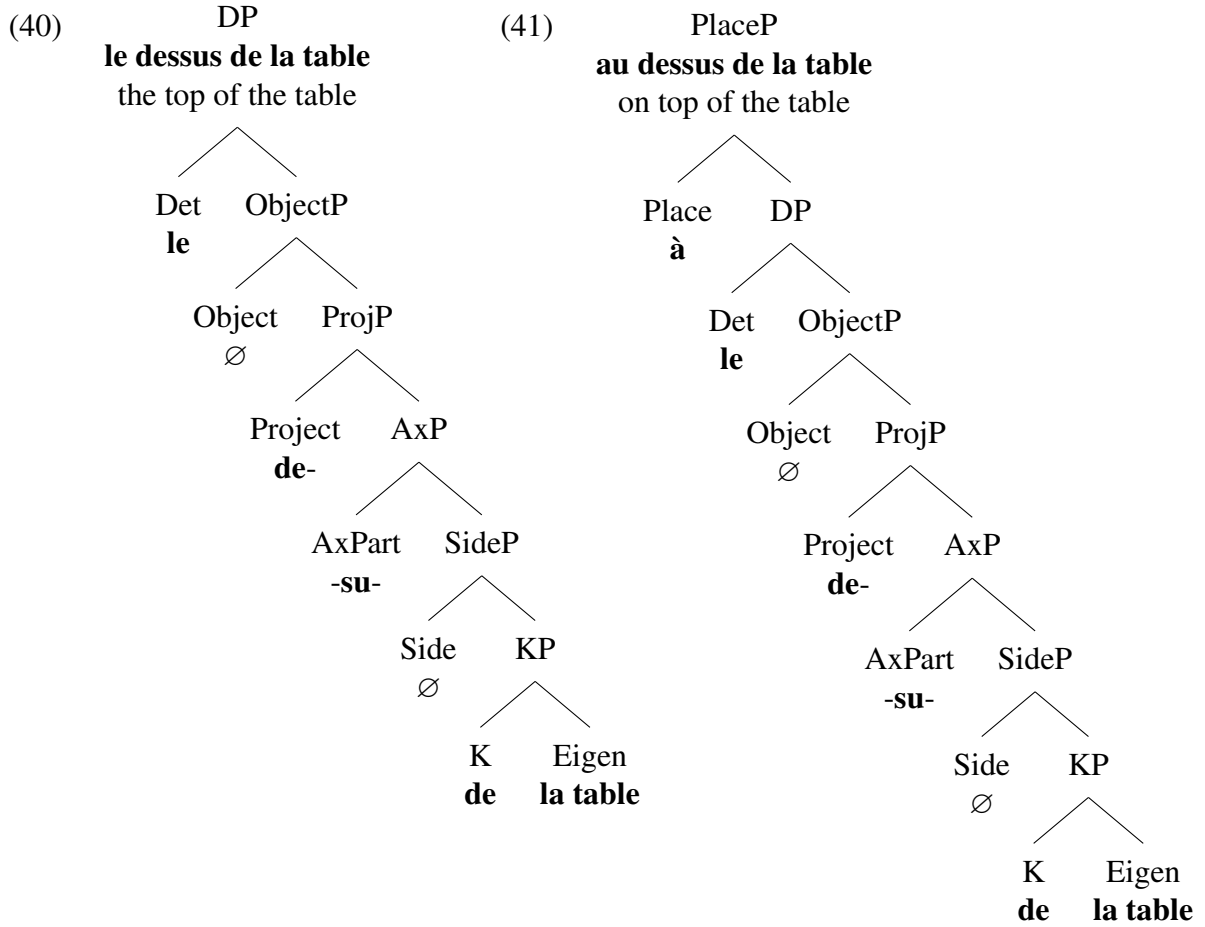
$$(38) \quad \begin{array}{ll} \text{a.} & \llbracket \text{dessous}_N \rrbracket = \lambda x. \text{object}(\text{bottom}(x)) \\ \text{b.} & \llbracket \text{dessous}_N \rrbracket = \lambda x. \text{object}(\text{project}_{1/2}(\text{bottom}(x))) \end{array}$$

$$(39) \quad \begin{array}{ll} \text{a.} & \llbracket \text{project}_1 \rrbracket = \lambda f \lambda u. \exists w [f(w) \wedge \text{start}(u) = \text{end}(w) \wedge \text{dir}(u) = \text{dir}(w)] \\ & \quad \quad \quad (\text{Matushansky \& Zwarts 2019: 275}) \\ \text{b.} & \llbracket \text{project}_2 \rrbracket = \lambda f \lambda u. \exists w [f(w) \wedge \text{start}(u) = \text{start}(w) \wedge \text{dir}(u) = (\text{dir}(w))] \end{array}$$

We prefer the second option for several reasons. Firstly, **object** is presumably unexpressed in other axial nominalizations that do not contain the prefix *de*-, like *l’avant de* ‘the front of’. We therefore need to assume that **object** is silent in other formations anyway. Secondly, assuming that *de*- is lexically ambiguous and can realize either (39a) or (39b), we can account for the three possible referents for *le dessous la table*:

1. the bottom of the table: generated via **project**₂ and **object**;
2. the region below the table: generated via **project**₁ and **object**;
3. both the bottom of the table and the region below it: generated via **project**₂ and **object**.

The tree structure we propose for axial nominalizations is given in (40), which contains Object and Det projections above ProjP. We take multi-word Ps such as *au dessus de* ‘above’ in *au dessus de la table*, to embed an axial nominalization, with a Place preposition (*à*, *en* or *par*) added on top, shown in (41). This returns a region located with respect to the entity referred by the axial nominalization.



We have now established that the *de-* prefix lexicalizes the functional element *project* on the basis of *de*-prefixed axial nominalizations. We next turn to the properties of the *de*-prefixed prepositional formations in French.

4. *de-* in locative prepositions

While the difference in the argument structure of simple locative Ps and complex Ps has been amply debated, there is surprisingly little discussion about the different morphological pieces in French complex prepositions, and *de-* specifically. Although Ruwet (1982) proposed early on that simple locative Ps are ‘compressed’ versions of more complex ones, he does not offer arguments for a decompositional analysis of *de*-prefixed Ps.¹⁶ In this section, we discuss the role of the *de-* prefix in locative prepositions, and argue along with Troberg (2021) and Heidinger & Troberg (to appear) that the intransitivity associated with the *de-* prefix is the result of pragmatic competition, including in superessive Ps.

4.1. Intransitivity and competition

We follow the general approach introduced by Zribi-Hertz (1986) and developed further by Authier (2016) and Heidinger & Troberg (to appear), among others, whereby appar-

¹⁶ “Peut-être [...] serait-il possible de décomposer *dessus*, *dedans*, etc. en deux morphèmes *de* + *sur*, *de* + *dans* [...] et une règle supprimerait *de* devant *sur*, *dans*, etc. dans certains contextes. Toutefois, je n’ai pas d’argument convaincant en faveur de cette idée.” (Ruwet 1982: 339)

ently intransitive *de*-prefixed Ps are in fact underlyingly transitive. Zribi-Hertz calls these intransitive Ps ‘orphan’ Ps to convey the idea they have lost their overt complement, which is (at least in some cases) compulsorily replaced in standard French by a silent anaphorical resumptive *pro*-form (with definite and non-human features).¹⁷ This distinguishes orphan Ps from stranded Ps in languages such as English, for which the empty position has the property of a trace. For Zribi-Hertz, all prepositions except those serving as case markers (e.g., *à* ‘at’, *de* ‘of/from’) can be orphan Ps, including *avec* ‘with’, *pour* ‘for’; *de*-prefixed Ps are special only in that the superficial absence of the complement goes along with a change in the morphology. However, Zribi-Hertz does not offer an explanation for why *de*- is sometimes added to Ps when they are used intransitively.¹⁸

Rizzi (1986: 519), in a footnote, offers an explanation of Zribi-Hertz’s generalization about which Ps allow *pro* in terms of number of syllables: monosyllabic Ps like *à*, *de*, etc. do not allow *pro*, while multisyllabic ones do. Rizzi speculates that monosyllabic Ps must cliticize into their complement in French; if the complement is not overtly realized, then cliticization is not possible. Prefixation with *de*- would thus make monosyllabic Ps like *dans*, *sur*, *sous* heavy enough to resist cliticization and thus license *pro*. An obvious problem for this suggestion, however, is that some monosyllabic Ps *do* allow intransitive uses in French, as illustrated in (42) for *sans* ‘without’ and *pour* ‘for’.¹⁹ A phonological explanation is thus insufficient to capture the distributional differences between simple and complex Ps.

- (42) a. Mon chapeau, je suis venu sans.
 my hat I am come without
 ‘I came without my hat.’ (Zribi-Hertz 1986: 627)
- b. La loi Veil, toutes les femmes étaient pour.
 the law Veil all the women were for
 ‘All women were for the Veil (abortion) Law.’ (Zribi-Hertz 1986: 632)

While other discussions of *de*-prefixation have occasionally surfaced more recently, we have found Troberg’s (2021) work on the *de*-/Ø alternation in French locative Ps based on their historical development to be particularly insightful. Troberg observes that in Medieval French, non-prefixed Ps were always transitive, while their *de*-prefixed counterparts were used both in transitive and intransitive frames (see also Rainsford 2019, Amiot et al. 2020 on the diachrony of these prepositions). At a later stage, *de*-prefixed Ps became specialized to intransitive contexts and occurred rarely in transitive ones. Troberg

¹⁷ Building on Gabriel (2002), Authier (2016) and Heidinger (2023, 2024), Heidinger & Troberg (to appear) distinguish three types of implicit arguments in French PPs on the basis of several tests (the availability of sloppy readings for the implicit argument and the presence of a contrastive focus on P): null pronouns, implicit arguments that are not syntactically projected, and ellipsis (as in (43a)).

¹⁸ A minority perspective holds that intransitive Ps, including those prefixed with *de*-, are truly intransitive, either because they are fully grammaticalized into adverbs, or because they semantically behave like relational definite descriptions linked to an overt or covert antecedent (as in *I have a new car. The handlebar is in leather*) (Olivier 2007). Again, however, such analyses do not make clear why *de*- appears on some but not all of these intransitive Ps.

¹⁹ As Zribi-Hertz (1986) observes, the divide between Ps that disallow vs. allow a covert complement rather seems to be between case-marking/functional Ps and the others (Cinque’s (2010) lexical Ps). But since *dans*, *sur*, *sous* on the one hand and *pour*, *sans* on the other are all lexical Ps, this generalization still does not account for why the former, but not the latter, need to be prefixed by *de*- to license *pro*.

A strong argument in favor of such a competition-based view is that *de*-prefixed Ps are strictly intransitive in standard French only when they have a non-prefixed competitor, like *(de)dans*, *(de)ssus*, *(de)ssous*. By contrast, *de*-prefixed Ps that have no morphologically simple counterparts, like *devant* and *derrière*, have both transitive and intransitive uses, as shown in (43).

- The fact that intransitivity depends on the presence of a morphologically simple competitor makes it unlikely that the *de*- prefix itself is inherently responsible for the intransitivity of the complex P. Rather, what Troberg’s reflections suggest is that the *de*-prefixed form in fact alternates between transitive and intransitive uses in the grammar. This accounts for its transitive uses in non-standard variants and when there is no competition. In standard French, however, the transitive use is blocked when a morphologically simpler counterpart is specialized in the transitive use. We propose that this blocking effect is pragmatic, and can be accounted for by a higher ranking of pragmatic constraints such as the Manner *Be Brief* submaxim in standard variants of natural languages. By contrast, non-standard variants such as colloquial or child French would be more lenient with such principles and rank higher competing principles such as *Be Expressive* (see Martin 2022, Martin et al. 2023), which requires a one-to-one mapping between form and meaning and thereby an overt spell-out of the ablative meaning in locative Ps.

This analysis allows us to maintain our proposal that *de-* is inherently an ablative marker which lexicalizes **project**₁ or **project**₂, rather than a marker of syntactic intransitivity. It also aligns with the idea that simple and complex locative Ps have the same conceptual structure. What differs between forms prefixed with *de-* or not is simply that **project** is overtly pronounced in the former case (e.g., *dessous* ‘from.under’) and silent in the latter (e.g., *sous* ‘under’).

4.2. Superessive prepositions

4.2.1. Data

In the proposed approach, simple locative Ps involve a **project** projection which happens to be silent. One potential counter-argument against the presence of **project** in simple Ps concerns the preposition *sur* ‘on’. It has frequently been pointed out across languages that among simple locative Ps, superessive Ps are special in that they seem to require physical contact with the object of reference; see, e.g., Svenonius 2010 and Gärdenfors 2020 on English *on*. This is illustrated by the contrast in (44) observed for French and English. In

(44a) with *au dessus de* ‘above’, the frame is not in direct contact with the shelf. In (44b), however, the use of *sur* conveys a requirement that the frame must be in contact with the shelf and receive support from it, which is strange in this context (Gärdenfors 2020).

- (44) a. Le cadre pend au dessus de l’étagère.
the frame hangs at.the from.on of the shelf
‘The frame hangs above the shelf.’
b. ?#Le cadre pend sur l’étagère.
the frame hangs on the shelf
‘#The frame hangs on the shelf.’

From these data it might be tempting to assume that only complex superessive Ps such as *au dessus de* involves *project*₁ or *project*₂, which returns a region projected from an axial part. The contact requirement of *sur* ‘on’ would be explained if ProjP was not part of its underlying conceptual structure.

However, we see several arguments in favor of a silent ProjP in the structure of *sur* ‘on’. Firstly, *sur* does license absence of contact in a few cases, as in (45) (see Vandeloise 1986a, Dendale & De Mulder 1997 for other attested examples).

- (45) a. Rachida a le nez sur son livre.
Rachida has the nose on her book
‘Rachida has her nose in her book.’
b. Les nuages sont sur nos têtes, l’orage est pour bientôt.
the clouds are on our head, the storm is for soon
‘The clouds are above our heads; the storm is coming soon.’ (Internet)

Second, contact is also typically required by the *de*-prefixed P *dessus* ‘from.on’. (47), for instance, is as strange as (44a), despite the presence of *de*- which overtly spells out a variant of *project*.²⁰

- (47)?#L’étagère, le cadre pend dessus.
the shelf the frame hangs from.on
‘The shelf, the frame hangs on it.’

Third, the transitive uses of *dessus* found in non-standard French also typically require contact, as shown in (48).

²⁰ Equivalent examples to (45) for *dessus* ‘from.on’ are also possible, as in (??).

- (46) Ton livre, j’ai le nez dessus.
your book I have the nose from-on
‘Your book, I’ve my nose in it.’

- (48) si jé té connaissais pas comme jé té connais, tu aurais déjà
 if I you knew NEG like I you know you would.have already
 ma main dessus la gueule.
 my hand from.on the face
 If I didn't know you like I know you, I'd have already slapped you.'

André Malraux, *Les Conquérants*

By contrast, *au dessus de* 'at from.on of', headed with the spatial preposition à 'at', often implicates the absence of contact with the object of reference (but does not entail it).

4.2.2. Analysis

Based on these data, we argue that at least in French and perhaps in other languages, superessive spatial Ps always involve ProjP, and, as a result, can always describe outwards regions projected from the top of an object, accounting for examples such as (45)-(??). That is, in the semantics, superessive Ps such as *sur*, *dessus* and *au dessus* can all express situations without contact, because they involve ProjP in their structure.

So why do superessive Ps convey contact between the Figure and Ground, if they always involve ProjP? We suggest that in superessive Ps, *project*₁ returns vectors whose length varies contextually.²¹ When they express contact between Figure and Ground, their length is close to zero. As a result, P ends up describing the surface of the top part of the object. By contrast, when they express absence of contact as in (45), *project*₁ returns vectors whose endpoint is situated outside the object of reference. Thus, both *sur* 'on' and *au dessus de* 'above' can express spatial configurations with or without contact between the Figure and the Ground.

But then why are *simple* superessive Ps typically understood to convey contact between the Figure and Ground? We once again propose that it is rather a matter of pragmatic division of labor which explains why *sur/ dessus* are preferred to express contact and *au dessus de/ au dessus* 'at.the from.on (of)' to express absence of contact. The reason for this is that à, like English *at*, conveys *proximity*: it expresses that the Figure is 'more closely related to a reference object than to any other in the [contrast] set' (Vasardani et al. 2017). By pragmatic strengthening, *proximity* with the object of reference is understood as absence of contact with it, and the simple superessive P gets specialized in the expression of contact.²²

Some evidence for this pragmatic account of the competition between *sur/ au dessus de* and *sous* 'under'/ *en dessous de* 'below' comes from Piérart's (1978) compre-

²¹ The reason why superessive Ps are special (in that the meaning of *project* can be modulated) is because of gravity: objects which are in the region projected from the top of an object typically have contact with this object.

²² See Svenonius (2010: 135) for an alternative account of the contact inference triggered by (*up*)on in English in terms of incompatibility with Deg_u (called "Meas" in Svenonius & Kennedy 2006), namely a Degree head that takes a measure phrase as its specifier. That (*up*)on is incompatible with measure phrases can be seen in (49).

(49) We were (*six feet) upon the trees.

(Svenonius 2010: 154)

Note that in French, measure phrases are unacceptable with *sur* but acceptable with *au/en dessus de*. This is true even if the distance between the Figure and the object of reference is close to zero, shown in (50).

hension study of these Ps in child French. Piérart asked children and adults to draw figures following verbal instructions that contained simple and complex Ps. Unlike adults, children up to the age of 6 drew relations involving *au dessus de* using connected figures. Piérart thus found that children did not seem to compute the inference of absence of contact triggered by the use of *au dessus de*. Implicatures, including manner implicatures, are known to be challenging for young children (Guasti et al. 2005, Wilson 2017, Wilson & Katsos 2020). If the non-contact inference is indeed an implicature obtained via the competition with the simpler form, as we suggest, then we can account for the relatively late development of comprehension of complex Ps in child French.

5. *de*-prefixed prepositions in child French

We have seen that in standard French, *de*-prefixed Ps competing with unprefixed ones cannot appear with an overt noun complement, while simple Ps without *de*- must appear with an overt complement. However, transitive uses of *de*-prefixed Ps have been observed in non-standard adult varieties of French and anecdotally in child French Pallaud & Savelli (2001). We conducted a corpus study on the use of simple vs. complex Ps in child French, which confirmed that young French learners do indeed overuse the *de*-prefixed locative Ps in transitive contexts compared to adults. That is, children often use complex Ps in contexts where adults use simple Ps. This behavior can be explained if simple and complex Ps share the same functional structure, and children choose to realize more of this structure overtly. Children's overuse of *de*- thus reflects their drive to pronounce the project function that usually remains silent in simple Ps.

5.1. Child French corpus study

Our corpus study investigates the use of complex Ps in French CHILDES (MacWhinney 2000). Using the CLAN program (MacWhinney, 2000), we extracted all occurrences of *dedans*, *dessous* and *dessus* and their conversational contexts in nine French CHILDES corpora. This yielded $N=1855$ utterances from 209 typically developing children aged 1 to 11, as shown as shown in Table 2.²³

We annotated for whether the *de*-prefixed preposition was used transitively (with an overt direct complement) or intransitively (without a complement). The overall results summarized in Table 3 show that children used the three *de*-prefixed complex prepositions with a complement on average 7% of the time. Some examples of this non-target use of complex Ps in child French are given in (51).

- (50) Celle-ci se met à danser dans tous les sens, comme en 'lévitation' à une
 the latter REFL puts to dance in all the directions like in levitation at a
 fraction de millimètre au-dessus de [/sur] la plaque métallique.
 fraction of millimeter at from.on of on the sheet metal
 'It starts dancing in all directions, as if in levitation, just a fraction of a millimeter above/on the metal plate.'
 (www.pourlascience.fr)

It is unclear to us what governs the French Ps' compability with measure phrases. At first sight, *dans* 'in', *face à* 'in front of', *au fond de* 'at the bottom of' are incompatible with them, while *sous* 'below', *vers* 'towards', *au dedans de* 'inside' accept them.

²³ The annotated child data are available at doi.org/10.5281/zenodo.17524424

Table 2. French CHILDES corpora with *dedans/dessous/dessus* utterances

Corpus	Children	
Goad-Rose	2	(Rose, 2000)
Leveillé	1	(Suppes et al., 1973)
Lyon	5	(Demuth & Tremblay, 2008)
MTLN	56	(Le Normand, 1986)
Palasis-1	12	(Palasis, 2010)
Palasis-2	16	(Palasis, 2010)
Paris	6	(Morgenstern & Parisse, 2007)
Vion-Colas	105	(Vion & Colas, 1998)
York	3	(De Cat & Plunkett, 2002)
Total	209	

Table 3. Use of *dedans/ dessus/ dessous* in child French in CHILDES corpora

	INTRANS	TRANS	TOTAL	% TRANS
dedans	1187	75	1262	5,9%
dessus	343	33	376	15,2%
dessous	195	22	217	8,0%
TOTAL	1725	130	1855	7,0%

- (51) a. je mets la voiture dedans la maison
 I put the car from.in the house
 ‘I’m putting the car into the house.’ (Philippe, 2;03, *Leveillé*)
- b. veux veux quoi papa dedans ton verre?
 want want what papa from.in your
 ‘What do you want, papa, in your glass?’ (Aurélie, 2;09, *MTLN*)
- c. c’est c’est c’est les gens ils dort dessus le parquet?
 it’s it’s it’s the people they sleeps from.on the floor
 ‘That’s, that’s the people, they sleep on the floor?’ (Anais, 2;05, *Lyon*)

To provide a baseline of comparison for our child data, we also examined child-directed speech in the Leveillé corpus. Of the nine French corpora examined, the Leveillé corpus provided the most data from a single child, Philippe; it also contained a large proportion of adult speech. We extracted all occurrences of *dedans*, *dessous* and *dessus* produced by adults in the Leveillé corpus and annotated their intransitive and transitive uses. The results in Table 4 show that transitive uses of complex Ps are very rare in

Table 4. Use of *dedans/ dessus/ dessous* in child and adult French in the Leveillé corpus

	INTRANS	TRANS	TOTAL	% TRANS
Philippe	209	16	225	7,1%
Adults	280	2	282	0,7%

child-directed speech, with only 2 occurrences, repeated in (52).

- (52) a. mais non, on va mettre le canard dedans le bateau, vous
 but no we will put the duck from.in the boat you
 comprenez rien.
 understand nothing
 ‘No, we’re putting the duck in the boat, you guys don’t understand.’
 (child-directed speech, *Leveillé*)
- b. Dedans quoi?
 from.in what
 ‘In what?’
 (child-directed speech, *Leveillé*)

Thus Philippe produced *de*-prefixed Ps with an overt complement at ten times the rate of the adults around him, at 7,1% compared to their 0,7%. The transitive uses of *de*-prefixed Ps reported for child French therefore cannot be attributed to language input but were innovated by the children themselves.

We gathered adult French data from several additional sources for further comparison with our child data. We collected all occurrences ($N=241$) of *dedans*, *dessus*, and *dessous* in the *Corpus du Français Parlé de nos Régions* (Delafontaine 2021), and found only a single occurrence built with an overt complement, involving *dedans* (53). Of the same prepositions ($N=87$) found in the *Corpus de Français Parlé à Bruxelles* (Labeau & Dister 2016), none occurred with an overt complement.

- (53) quand on en a après quelqu’un on prend un caillou on va le
 when one PART.PRN has after someone one takes a pebble one goes it
 lui mettre dedans la tronche là oui
 them put from.in the mug there yes
 ‘When you’re mad at someone, you just grab a rock and go smash it right in their face, yeah.’

Finally, we computed the proportion of occurrences of *dedans/ dessus/ dessous* preceded by a verb and followed by a determiner in the set of occurrences of *(de)dans/ (de)ssus/ (de)ssous* in the same syntactic context in the FrTenTen17 web corpus via Sketch-Engine (Kilgarriff et al. 2014). While this is a written corpus, it contains informal genres such as blogs or forums. As shown in Table 5, locative prepositions occur much less frequently with the preposition *de* in such transitive frames than in our child data (less than 0.1% of the time).²⁴ From this wide range of adult data, we can confirm that *dedans*, *dessous* and *dessus* are found with an overt complement significantly more often in child French than in adult French.

²⁴ The percentage is likely lower in view of the fact that many tokens are false positives, where the *de*-P is a target-like intransitive P (followed by the direct object of the verb for instance), or even not a complement of the preceding verb. Furthermore, some of the very few transitive uses of *de*-Ps in this dataset are not written in contemporary French, but pertain to a previous stage of the language.

Table 5. Use of *(de)dans/ (de)ssus/ (de)ssous* preceded by a verb and followed by a definite overt complement in adult French on FrTenTen17 (Kilgariff et al. 2014)

String	Number of hits	% in the total (with and without <i>de</i>)
V+dans DET	13.719.091	
V+dedans DET	3.087	0,02%
V+sur DET	8.730.039	
V+dessus+DET	7.703	0,08%
V+sous DET	1.039.244	
V+dessous DET	758	0,07%

5.2. Analysis

The overuse of *de-* in transitive frames in child French can be accounted for in two different ways. Adopting Zribi-Hertz's (1986) perspective, overuse of *de-* can be seen as a tendency to realize an overt form where one expects silent *pro*. In this view, the overuse of *de-* in locative Ps results from children's tendency to realize *pro* overtly. A prediction of such an account is that children should also prefer to overtly realize the complement of other orphan Ps, including those which are not prefixed with *de-*, such as *avec* 'with', etc. While fully testing this prediction would require additional corpus studies, what we find anecdotally is that French children in fact overuse orphan Ps *without* complements. For instance, children use orphan Ps not only in contexts where *pro* has a non-human feature (like adults), but also in contexts where it has a human feature, as in (54). This suggests that children need time to acquire the selectional constraints of orphan Ps, since orphan Ps with a covert *animate* complement are non-standard in French (Zribi-Hertz 1986: 603), although restrictions in terms of animacy have been shown to be stronger in non-contrastive than in contrastive contexts (Heidinger 2024).

(54) %Papa vient avec.
 papa comes with
 'Papa comes with me/us/you/them.'

A second explanation of the child data takes advantage of the view developed here that simple and complex Ps have the same functional structure. This means that both the simple and *de*-prefixed Ps are syntactically available as transitive Ps. Standard adult French, however, bans the *de*-prefixed P when it competes with a simple P because of a pragmatic preference for more economical forms. What French children are doing when overusing *de-*, then, is overtly realizing a piece of conceptual structure which is silent in the adult language, overriding economy constraints. In morphologically simple Ps, there is a mismatch between morphology and conceptual structure, as they express the same complex conceptual structure realized by morphologically complex Ps. Such mismatches are challenging for children, who are known to more easily acquire one-to-one form-to-meaning mappings (Slobin 1973, van Hout 2008, Martin et al. 2020, Guasti et al. 2023).

We have proposed in previous sections that adult speakers of standard French *ceteris paribus* tend to choose the most economical P form, ranking pragmatic constraints

like *Be Brief* over *Be Expressive*. Children, by contrast, are still developing the optimal trade-off between minimizing production effort and maximizing understandability (Tal & Arnon 2019, 2022), which sometimes results in the production of too few morphemes (underuse or omission errors) and sometimes too many morphemes (overuse or commission errors). Research on child language within the Meaning First Approach to the mapping between conceptual structure and morphology (Sauerland & Alexiadou 2020) has addressed this tension explicitly. Following Martin et al. (2023) and Alexiadou (2024), we suggest that while adults prioritize Siddiqi's (2006) *Minimize Exponence* general economy principle over the opposite *Maximize Exponence* transparency principle, the latter is ranker higher for children than it is for adults. In order to obey *Maximize Exponence*, children may realize more complex prepositional forms.

- (55) a. *Minimize Exponence*:
Realize a set of concepts using the fewest exponents.
b. *Maximize Exponence*:
Realize each concept using exactly one exponent.

In favor of an account along these lines, we note other examples in which children overtly express a piece of conceptual structure that is usually silent in the target language. For instance, German children occasionally overtly spell out the functional concept WITH typically silent in the caritive preposition *ohne* 'without' (Sauerland et al. 2024; see van Kampen 2024 for an alternative account of this pattern).

- (56) %Die Katze mit ohne Mütze (child German)
the cat with without hat
'The cat without a hat.' (target: *Die Katze ohne Mütze*)

Additionally, Italian children tend to overuse multi-word Ps and in particular the genitive case marker *a* in the production of prepositions (Dal Farra et al. 2022), as in (57):

- (57) %Dentro a quel buco c'era un gufo. (Giorga, 8)
inside at that hole there.was an owl
'Inside that hole there was an owl.' (target: *In quel buco c'era un gufo*)

In French, children tend to mark overt genitive or dative semantics via a dedicated case marking (functional) preposition rather than via possessives or clitics, as shown in (58)-(59).

- (58) a. dans mon cou (adult French)
in my neck
b. %dans le cou de moi (child French)
in the neck of me

- (59) a. Tu me l'as donné. (adult French)
 you CL-DAT.1SG CL-ACC.3SG has given
- b. %Tu l'as donné à moi. (child French)
 you CL-ACC.3SG has given to me

Finally, in child Romanian and child Spanish, Ticio & Avram (2015) 2015 note overextensions of DOM marking in child Romanian and child Spanish. We thus observe a more general tendency of children to overtly express functional material in a one-to-one fashion with complex Ps. French children's non-target-like overt exponence of *de*-instantiates a more general cross-linguistic pattern in acquisition: they overtly express the project function, a core ingredient of locative Ps in French as well as other languages.

6. Conclusion

In this paper, we argued that in *de*-prefixed spatial Ps or axial nominalizations, *de*- is an ablative (Source) marker lexicalizing the project function. This explains its surprising presence in locative prepositions. As language acquisition feeds into language change (Kroch 2005, Lightfoot 2010, Cournane 2017), the overuse of *de*- by L1 French learners may be seen as a return to the initial stage of a locative prepositional cycle. If children do indeed go back to the beginning of the cycle, this has an interesting consequence for processes of language change in general, namely that conceptual structure never truly disappears with the natural processes of morphological erosion.

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